



Yakeen Plus (2025)

SHORT PRACTICE TEST - 03

DURATION: 60 Minutes

DATE: 25/09/2024

M. MARKS: 192

Topics Covered

Physics:	Work, energy and power, Centre of mass and System of Particles, Rotational Motion
Chemistry:	Thermodynamics and Thermochemistry, Chemical Equilibrium Ionic Equilibrium
Biology:	(Botany) : The Living World (Zoology) : Body Fluids and Circulation

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **60 minutes** duration.
3. The test booklet consists of **48** questions. The maximum marks are **192**.
4. All questions are compulsory.
5. There is only one correct response for each question.
6. Each correct answer will give **4** marks while **1** Mark will be deducted for a wrong MCQ response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.
9. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS) : _____

Roll Number : _____

OMR Bar Code Number : _____

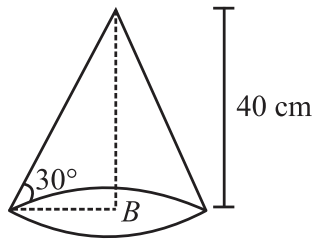
Candidate's Signature : _____ **Invigilator's Signature** _____

SECTION-(I) PHYSICS

1. A stone of mass 0.2 kg is tied to one end of a thread of length 0.1 m whirled in a vertical circle. When the stone is at the lowest point of circle, tension in thread is 52N, then velocity of the stone will be:

(1) 4 m/s (2) 5 m/s
(3) 6 m/s (4) 7 m/s

2. A uniform solid cone of height 40 cm is shown in figure. The distance of centre of mass of the cone from point B (Centre of the base) is:



(1) 20 cm (2) 10/3 cm
(3) 20/3 cm (4) 10 cm

3. Two bodies with masses m_1 and m_2 ($m_1 > m_2$) are joined by a string passing over fixed pulley. Assume masses of the pulley and thread negligible. Then the acceleration of the centre of mass:

(1) $\left(\frac{m_1 - m_2}{m_1 + m_2}\right)g$ (2) $\left(\frac{m_1 - m_2}{m_1 + m_2}\right)^2 g$
(3) $\frac{m_1 g}{m_1 + m_2}$ (4) $\frac{m_2 g}{m_1 + m_2}$

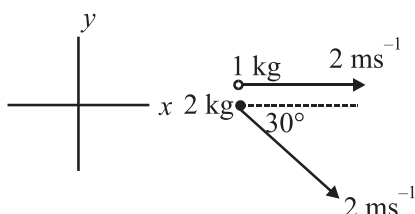
4. If we throw a body upwards with velocity of 4 m/s, at what height does its kinetic energy reduce to half of the initial value? (Take $g = 10 \text{ m s}^{-2}$)

(1) 4 m (2) 2 m
(3) 1 m (4) 0.4 m

5. Two springs A and B ($k_A = 2k_B$) are stretched by applying forces of equal magnitudes at the four ends. If the energy stored in A is E, that in B is

(1) E/2 (2) 2E
(3) E (4) E/4

6. Find the velocity of centre of mass of the system shown in the figure.



(1) $\left(\frac{2+2\sqrt{3}}{3}\right)\hat{i} - \frac{2}{3}\hat{j}$

(2) $4\hat{j}$

(3) $\left(\frac{2-2\sqrt{3}}{3}\right)\hat{i} - \frac{2}{3}\hat{j}$

(4) None of these

7. A light rod of length l has two masses m_1 and m_2 attached to its two ends. The moment of inertia of the system about an axis perpendicular to the rod and passing through the centre of mass is:

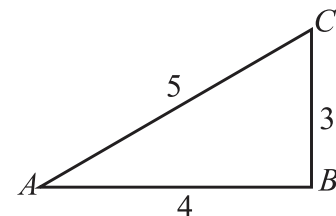
(1) $\frac{m_1 m_2}{m_1 + m_2} l^2$

(2) $\frac{m_1 + m_2}{m_1 m_2} l^2$

(3) $(m_1 + m_2) l^2$

(4) $\sqrt{m_1 m_2} l^2$

8. The ABC is a triangular plate of uniform thickness. The sides are in the ratio shown in the figure. I_{AB} , I_{BC} and I_{CA} are the moments of inertia of the plate about AB, BC and CA respectively. Which one of the following relations is **correct**?



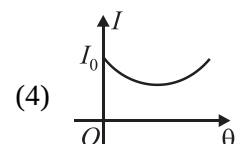
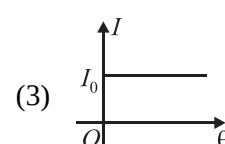
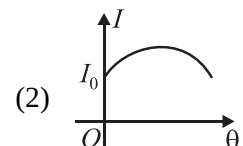
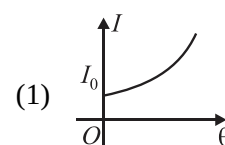
(1) $I_{AB} + I_{BC} = I_{CA}$

(2) I_{CA} is maximum

(3) $I_{AB} > I_{BC}$

(4) $I_{BC} > I_{AB}$

9. A square plate has a moment of inertia I_0 about an axis lying in its plane, passing through its centre and making an angle θ with one of the sides. Which graph represents the variation of I with θ ?



10. A person of mass m is standing on one end of a plank of mass M and length L and floating in water. The person moves from one end to another and stops. The displacement of the plank is:

- (1) $\frac{Lm}{(m+M)}$
- (2) $Lm(M+m)$
- (3) $L \frac{(m+M)}{m}$
- (4) $\frac{LM}{(m+M)}$

11. If $\vec{F}$ is the force acting on a particle having position vector $\vec{r}$ and $\vec{\tau}$ be the torque of this force about the origin, then:-

- (1) $\vec{r} \cdot \vec{\tau} = 0$ and $\vec{F} \cdot \vec{\tau} \neq 0$
- (2) $\vec{r} \cdot \vec{\tau} \neq 0$ and $\vec{F} \cdot \vec{\tau} = 0$
- (3) $\vec{r} \cdot \vec{\tau} > 0$ and $\vec{F} \cdot \vec{\tau} < 0$
- (4) $\vec{r} \cdot \vec{\tau} = 0$ and $\vec{F} \cdot \vec{\tau} = 0$

12. The moment of inertia of a disc of radius 0.5 m about its geometric axis is $2\text{kg}\cdot\text{m}^2$. If a string is tied to its circumference and a force of 10 Newton is applied, the value of torque with respect to this axis will be:

- (1) 2.5 N-m
- (2) 5 N-m
- (3) 10 N-m
- (4) 20 N-m

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Thermodynamics and Thermochemistry, Chemical Equilibrium Ionic Equilibrium

SECTION-(II) CHEMISTRY

13. Ratio of active masses of 22 g CO_2 , 3 g H_2 and 7 g N_2 in a gaseous mixture:

- (1) 22 : 3 : 7
- (2) 0.5 : 3 : 7
- (3) 1 : 3 : 1
- (4) 1 : 3 : 0.5

14. Match List-I with List-II.

List-I (Hypothetical reactions)		List-II $\left(\frac{K_p}{K_c} \text{ ratio}\right)$	
(A)	$\text{A}_2(\text{g}) + 3\text{B}_2(\text{g}) \rightleftharpoons 2\text{AB}_3(\text{g})$	(I)	$(RT)^{-2}$
(B)	$\text{A}_2(\text{g}) + \text{B}_2(\text{g}) \rightleftharpoons 2\text{AB}(\text{g})$	(II)	$(RT)^0$
(C)	$\text{A}(\text{s}) + 1.5\text{B}_2(\text{g}) \rightleftharpoons \text{AB}_3(\text{g})$	(III)	$(RT)^{1/2}$
(D)	$\text{AB}_2(\text{g}) \rightleftharpoons \text{AB}(\text{g}) + 0.5\text{B}_2(\text{g})$	(IV)	$(RT)^{-1/2}$

Choose the **correct** answer from the options given below:

- (1) A-I, B-II, C-III, D-IV
- (2) A-II, B-I, C-IV, D-III
- (3) A-I, B-IV, C-III, D-II
- (4) A-I, B-II, C-IV, D-III

15. Which of the following is **not** a state function?

- (1) Internal energy
- (2) Enthalpy
- (3) Work
- (4) Entropy

16. In a closed insulated container, a liquid is stirred with a paddle to increase its temperature. In this process, which of the following is **true**?

- (1) $\Delta U = w = q = 0$
- (2) $\Delta U \neq 0, q = w = 0$
- (3) $\Delta U = w \neq 0, q = 0$
- (4) $\Delta U = q \neq 0, w = 0$

17. Given below are two statements:

Statement I: The gas phase reaction $\text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g}) \rightleftharpoons \text{PCl}_5(\text{g})$ shifts to the right on increasing pressure.

Statement II: When pressure increase, equilibrium shifts towards more number of moles. In the light of the above statements, choose the most appropriate answer from the options given below.

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

18. If w_1, w_2, w_3 and w_4 for an ideal gas are magnitude of work done in isothermal, adiabatic, isobaric and isochoric reversible expansion processes, the **correct** order will be:

- (1) $w_1 > w_2 > w_3 > w_4$
- (2) $w_3 > w_2 > w_1 > w_4$
- (3) $w_3 > w_2 > w_4 > w_1$
- (4) $w_3 > w_1 > w_2 > w_4$

19. 2 mole of an ideal gas at 27°C expands isothermally and reversibly from a volume of 4 litre to 40 litre. The work done (in kJ) by the gas is:
 (1) $w = -28.72$ kJ
 (2) $w = -11.488$ kJ
 (3) $w = -5.736$ kJ
 (4) $w = -4.988$ kJ
20. The value of $\Delta_f H^\circ$ for PCl_5 is
 Given: $\text{P}_4(\text{s}) + 6\text{Cl}_2(\text{g}) \rightarrow 4\text{PCl}_3(\text{l}); \Delta H = -1270$ kJ
 $\text{PCl}_3(\text{l}) + \text{Cl}_2(\text{g}) \rightarrow \text{PCl}_5(\text{s}); \Delta H = -137$ kJ
 (1) 772 kJ/mol (2) -498 kJ/mol
 (3) -454.5 kJ/mol (4) -772 kJ/mol
21. The equilibrium constant for the reaction
 $\text{Br}_2 \rightleftharpoons 2\text{Br}$ at 500K and 700K are 1×10^{-10} and 1×10^{-5} respectively. The reaction is:
 (1) endothermic. (2) exothermic.
 (3) fast. (4) slow.
22. The solubility product of CuS , Ag_2S and HgS are 10^{-37} , 10^{-44} and 10^{-54} respectively. The solubility of these sulphides will be in the order:

- (1) $\text{HgS} > \text{Ag}_2\text{S} > \text{CuS}$
 (2) $\text{Ag}_2\text{S} > \text{HgS} > \text{CuS}$
 (3) $\text{CuS} > \text{Ag}_2\text{S} > \text{HgS}$
 (4) $\text{Ag}_2\text{S} > \text{CuS} > \text{HgS}$

23. The temperature of an ideal gas increases in an:
 (1) adiabatic expansion.
 (2) isothermal expansion.
 (3) adiabatic compression.
 (4) isothermal compression.
24. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: If water is heated to 50°C then pH will increase.
Reason R: K_w increase with increase in temperature.
 In the light of the above statements, choose the **correct** answer from the options given below.
 (1) A is true but R is false.
 (2) A is false but R is true.
 (3) Both A and R are true and R is the correct explanation of A.
 (4) Both A and R are true but R is not the correct explanation of A.

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 The Living World

SECTION-(III) BOTANY

25. Rice, cereals, monocots and plants represent:
 (1) different taxa at different level.
 (2) same taxa of different category.
 (3) different category of same taxa.
 (4) same category for different taxa.
26. Potato and brinjal belong to the genus *Solanum*, which reflects that:
 (1) they belong to single species.
 (2) they are a group of related species.
 (3) they both are morphologically and structurally similar to each other in all respects.
 (4) they can always produce fertile hybrid.
27. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Systematics is the study of organisms, their diversities but not the relationships among them.
Reason R: The word systematics is derived from the Latin word 'systema'.
 In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
 (2) A is false but R is true.
 (3) Both A and R are true and R is the correct explanation of A.
 (4) Both A and R are true but R is NOT the correct explanation of A.
28. Choose **incorrect** option w.r.t. *Panthera leo*.
 (1) Common name of tiger.
 (2) *Panthera* represents generic name.
 (3) *leo* represents specific epithet.
 (4) *Panthera* represents higher level of taxon than *leo*.

29. Match **List-I** with **List-II**.

List-I		List-II	
(A)	<i>Triticum aestivum</i>	(I)	Housefly
(B)	<i>Mangifera indica</i>	(II)	Leopard
(C)	<i>Musca domestica</i>	(III)	Wheat
(D)	<i>Panthera pardus</i>	(IV)	Mango

Choose the **correct** answer from the options given below:

- (1) A-III, B-IV, C-I, D-II
 (2) A-IV, B-I, C-III, D-II
 (3) A-III, B-I, C-II, D-IV
 (4) A-IV, B-III, C-I, D-II

30. Consider the following taxa and select how many of them do **not** belong to the lowest category of taxonomic hierarchy?

Chilli, Lion, Dicots, Animals, Mango, Sweet potato, Polymoniales, Cat

- (1) Seven (2) Six
(3) Five (4) Three

31. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Lower the taxa, greater is the difficulty of determining the relationship to other taxa at the same level.

Reason R: As we go higher from species to kingdom, the number of common characteristics goes on decreasing.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

32. Which of the following groups belong to the same taxonomic category?

- (1) Diptera, Hominidae, Poaceae
(2) Diptera, Sapindales, Poaceae
(3) Diptera, Sapindales, Poales
(4) Diptera, Hominidae, Poales

33. Read the following and identify the **correct** statements related to binomial nomenclature.

- A. Biological names are generally Italian in origin.
B. Both the words in a biological name, when handwritten, are separately underlined.
C. The genus name starts with a capital letter.
D. The specific epithet starts with a small letter.
E. Name of the author appears before the specific epithet.
F. Nomenclature is only possible when the organism is described correctly.

- (1) Only A, B, and C
(2) Only B, C, D and F
(3) Only A, C, D and E
(4) Only A, D and E

34. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Sapindales	(I)	Division
(B)	Muscidae	(II)	Order
(C)	<i>Mangifera</i>	(III)	Genus
(D)	Angiospermae	(IV)	Family

Choose the **correct** answer from the options given below:

- (1) A-II, B-IV, C-III, D-I
(2) A-II, B-III, C-IV, D-I
(3) A-I, B-IV, C-II, D-III
(4) A-I, B-III, C-IV, D-II

35. Choose the **incorrect** statement from the following.

- (1) *Homo* belongs to the order primata.
(2) Monkey, gorilla and gibbon are placed under a common group called Mammalia.
(3) Tiger, cat and dog are placed in a common group called Mammalia.
(4) Insecta is placed under phylum Chordata.

36. Read the following statements.

- A. As we go from species to kingdom in a taxonomic hierarchy, the number of organisms will decrease.
B. If two plants belong to the same division but in different orders, they may belong to the same class.
C. Families among the plant species are characterised on the basis of reproductive features only.
D. In case of plants, classes with a few similar characters are assigned to a higher category called phylum.
E. All animals belonging to various phyla are assigned to the highest category called kingdom Animalia.

How many is/are **correct**?

- (1) Only A and B
(2) Only C and D
(3) Only D and E
(4) Only B and E

SECTION-(IV) ZOOLOGY

37. Choose the **correct** option w.r.t. incomplete double circulation.

- (1) Fishes. (2) Amphibian.
(3) Mammals. (4) Birds.

38. Given below are two statements:

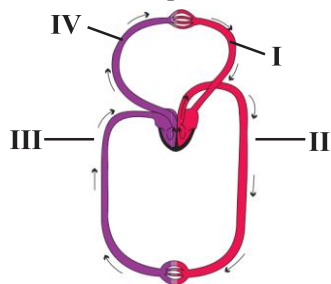
Statement I: In human heart the opening between right atrium and right ventricle is guarded by tricuspid valve.

Statement II: Pericardium is a double membrane wall which enclose the pericardial fluid and protect the heart.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

39. The given diagram shows schematic plan of blood circulation in human with labelled **I, II, III** and **IV**. Identify the labelled part along with their functions and select the **correct** option.



(1)	I	Pulmonary vein - takes impure blood from body part to heart.
(2)	II	Dorsal aorta - Carry blood from heart to body part.
(3)	III	Vena cava - carry blood from body part to left atrium.
(4)	IV	Pulmonary artery - takes blood from left auricle to lungs

40. Given below are two statements:

Statement I: Haemoglobin is a red coloured, iron containing complex protein.

Statement II: A healthy individual has 12-16 gms of haemoglobin in every 1000 ml of blood.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

41. Match **List-I** and **List-II**.

List-I		List-II	
(A)	Erythrocytes	(I)	6000-8000 mm ⁻³ of blood.
(B)	Leucocytes	(II)	60-65 per cent of total WBCs
(C)	Thrombocytes	(III)	5 to 5.5 millions mm ⁻³ of blood
(D)	Neutrophils	(IV)	1,500,00-3,500,00 mm ⁻³ of blood

Choose the **correct** answer from the options given below:

- (1) A-III, B-I, C-IV, D-II
(2) A-II, B-I, C-IV, D-III
(3) A-III, B-IV, C-II, D-I
(4) A-IV, B-I, C-III, D-II

42. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Lymph helps in the absorption of fat in the lacteals of intestinal villi.

Reason R: Lymph is a red coloured fluid containing specialised lymphocytes which are responsible for the immune responses of the body.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

43. The blood group which lacks antigen on the surface of RBCs is **X** and human with this blood group can receive blood from donors whose blood group is **Y**.

Choose the option that fill the blank **correctly**.

- (1) X-O, Y-O
(2) X-O, Y-AB
(3) X-AB, Y-O
(4) X-AB, Y-AB

44. Read the following statements (**A-E**) and select the **correct** option.

- A. Basophils secrete histamine, serotonin and heparin.
- B. Red blood cells are the most abundant cell of blood.
- C. Eosinophils resist infection and are also associated with allergic reaction.
- D. Lymphocytes are agranulocyte.
- E. Thrombocytes are cell fragments produced from megakaryocytes.

Choose the most appropriate answer from the options given below:

- (1) B, C, D, and E only
- (2) A, B and C only
- (3) B, D and E only
- (4) A, B, C, D and E

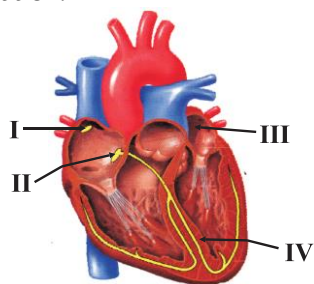
45. Match **List-I** and **List-II**.

List-I		List-II	
(A)	P wave	(I)	Beginning of ventricular systole
(B)	Q wave	(II)	Repolarisation of the ventricles
(C)	QRS complex	(III)	Depolarisation of ventricles
(D)	T wave	(IV)	Depolarisation of atria

Choose the **correct** answer from the options given below:

- (1) A-III, B-IV, C-II, D-I
- (2) A-I, B-IV, C-II, D-III
- (3) A-IV, B-I, C-III, D-II
- (4) A-III, B-I, C-IV, D-II

46. The diagram given below shows a section of a human heart with labelled **I, II, III** and **IV**. Select the **correct** option which gives correct identification.



(1)	I	Sino-atrial node present in the right upper corner of the left atrium.
(2)	II	Atrio-ventricular node present in the upper left corner of the right atrium.
(3)	III	Right atrium receive oxygenated blood.
(4)	IV	Thick-walled, inter-ventricular septum, separates the left and the right ventricles.

47. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Human body has ability to alter the stroke volume.

Reason R: Stroke volume is the volume blood which is pump out by both the atrium during cardiac cycle.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

48. _____ is a unique vascular connection between the digestive tract and liver.

Choose the option that fill the blank **correctly**.

- (1) Hepatic portal system.
- (2) Renal portal system.
- (3) Pulmonary circulation.
- (4) Systemic circulation.

